TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
THAPATHALI CAMPUS

A Project Report

On

Hierarchical Network Design

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Contents

1. INTRODUCTION

1.1. LAN:

A Local Area Network (LAN) is a network that connects computers and devices within a limited geographical area, such as a home, office, or building. LANs are designed to facilitate communication, resource sharing, and data exchange among connected devices. The key components of a LAN are:

i. **Routers:** Routers connect the LAN to other networks, such as the Internet. They route data between the LAN and external networks, using IP addresses to determine the best path for data packets.

ii. **Switches:** Switches are devices that connect multiple devices on a LAN. They use MAC addresses to forward data only to the intended recipient, reducing unnecessary traffic and improving network efficiency.

iii. **Cables and Connectors:** For wired LANs, Ethernet cables (such as Cat5e, Cat6) and connectors (RJ45) are used to physically connect devices to switches and routers.

iv. **Clients:** Clients are the end-user devices that connect to the LAN to access resources and services. Examples include computers, laptops, smartphones, and tablets.

1.2. VLAN:

Virtual Local Area Networks (VLANs) are a network technology that allows for the segmentation of a physical network into multiple logical networks, thereby enhancing security, performance, and manageability. By using unique VLAN IDs and managed switches, network administrators can isolate sensitive data, reduce broadcast traffic, and group devices based on function rather than physical location. VLAN tagging (IEEE 802.1Q) is employed to indicate the VLAN membership of data packets, enabling efficient traffic separation and management. This setup allows for both access ports, which connect devices to a single VLAN, and trunk ports, which carry traffic for multiple VLANs between switches. VLANs are particularly useful for departmental separation, guest network access, and ensuring quality of service for applications like VoIP, making them a crucial component in modern network design.

1.3. Inter-VLAN Routing:

Inter-VLAN routing is a crucial networking process that allows communication between devices in different VLANs within the same network, overcoming the isolation inherent to VLANs. This is achieved either through a router-on-a-stick configuration, where a single physical router interface is subdivided into multiple sub-interfaces, each tagged with a different VLAN ID, or by using a Layer 3 switch, which can route traffic directly between VLANs without needing a separate router. In both cases, IP routing is configured to enable data packets to move between VLANs, ensuring seamless communication and resource sharing across the network.

1.4. DHCP

Dynamic Host Configuration Protocol (DHCP) is a network management protocol that automates the assignment of IP addresses and other essential network configurations, such as subnet masks, default gateways, and DNS servers, to devices on a network. By dynamically distributing these settings, DHCP eliminates the need for manual configuration, reducing administrative workload and minimizing configuration errors. When a device connects to a network, it sends a DHCP

Discover message to locate available DHCP servers, which respond with an IP address offer. The device then requests the offered address, and the server confirms the lease with an acknowledgment, enabling seamless and efficient network connectivity.

1.5.RIP

RIP (Routing Information Protocol) is a dynamic routing protocol that automatically adjusts routing paths based on changes in network topology. Unlike static routing, which requires manual updates, RIP enables routers to dynamically discover and maintain routing information by periodically exchanging routing tables with neighboring routers. This allows RIP to adapt to changes, such as new or failed routes, and helps in finding the most efficient paths for data packets. RIP uses hop count as its metric, with a maximum limit of 15 hops, making it suitable for smaller networks.

2. NETWORK DESIGN

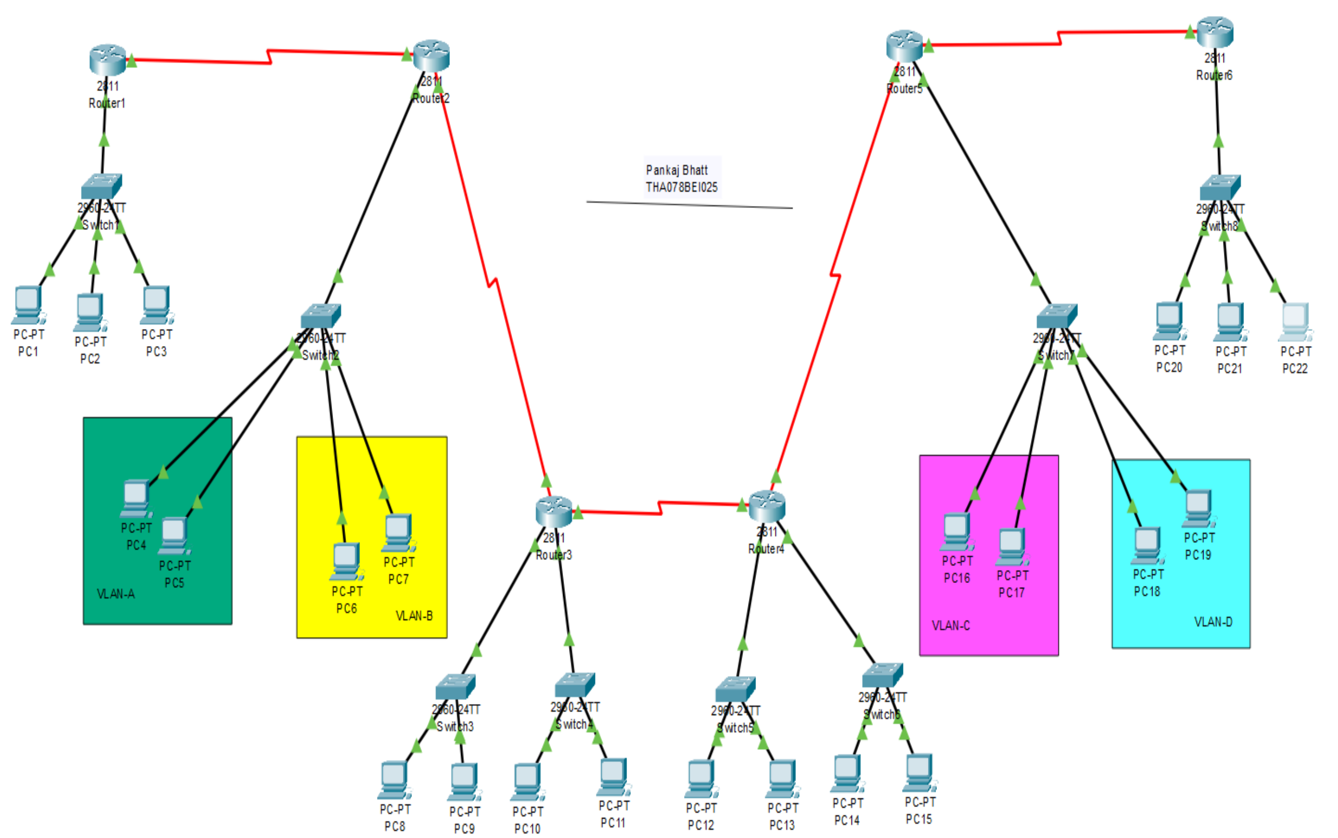


Figure: Design of the Network Implemented

This network design illustrates a complex, hierarchical topology with multiple interconnected routers and switches, effectively segmenting the network using VLANs for enhanced traffic management and security. Routers are interconnected to form the primary routing paths, while switches connect to PCs and organize them into different VLANs, such as VLAN-A, VLAN-B, VLAN-C, and VLAN-D. This setup supports inter-VLAN routing, allowing devices in different VLANs to communicate through the routers. The design offers scalability, allowing for the addition of more devices and VLANs, and ensures redundancy for network resilience. Overall, it demonstrates a well-structured network capable of efficient data flow and robust management.

3. LIST OF NETWORKS USED

- **VLAN 2 (VLAN-A)**: 192.168.2.0/24
- **VLAN 3 (VLAN-B)**: 192.168.3.0/24
- **VLAN 4 (VLAN-C)**: 192.168.4.0/24
- **VLAN 5 (VLAN-D)**: 192.168.5.0/24
- **Router 1 DHCP**: 192.168.80.0/24
- **Router 6 DHCP**: 192.168.90.0/24
- **Router 3 DHCP**: 192.168.40.0/24, 192.168.50.0/24
- **Router 4 DHCP**: 192.168.60.0/24, 192.168.70.0/24
- **RIP Networks**:

- 192.169.1.0/30
- 192.169.2.0/30
- 192.169.3.0/30
- 192.169.4.0/30
- 192.169.5.0/30

4. CREATING THE NETWORK

4.1. Creating VLANs:

4.1.1. Creating VLANs on Switch 2

Step 1: Creating VLANs

```
Switch>enable
Switch#configure terminal
Switch(config)#vlan 2
Switch(config-vlan)#name VLAN-A
Switch(config-vlan)#exit

Switch(config)#vlan 3
Switch(config-vlan)#name VLAN-B
Switch(config-vlan)#exit
```

Step 2: Assigning ports to VLANs

```
Switch(config)#interface range fastethernet 0/2-3
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 2
Switch(config-if-range)#exit

Switch(config)#interface range fastethernet 0/4-5
```

Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 3
Switch(config-if-range)#exit

Step 3: Trunking ports for Inter-VLAN Routing
Switch> enable
Switch# configure terminal
Switch(config)# interface fastethernet 0/1
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 2,3
Switch(config-if)# exit
Switch(config)# exit

Step 4: Configuring Inter-VLAN Routing configuration on Router 2
Router>enable
Router#configure terminal
Router(config)#interface fastethernet 0/0.2
Router(config-subif)# encapsulation dot1Q 2
Router(config-subif)#ip address 192.168.2.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface fastethernet 0/0.3
Router(config-subif)#encapsulation dot1Q 3
Router(config-subif)#ip address 192.168.3.1 255.255.255.0
Router(config-subif)#exit
Router(config)#exit

4.1.2. Creating VLANS on Switch 7

Step 1: Creating VLANs
Switch>enable
Switch#configure terminal
Switch(config)#vlan 4
Switch(config-vlan)#name VLAN-C
Switch(config-vlan)#exit

Switch(config)#vlan 5
Switch(config-vlan)#name VLAN-D
Switch(config-vlan)#exit

Step 2: Assigning ports to VLANs
Switch(config)#interface range fastethernet 0/1-2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 4
Switch(config-if-range)#exit

Switch(config)#interface range fastethernet 0/3-4

Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 5
Switch(config-if-range)#exit

Step 3: Trunking ports for Inter-VLAN Routing
Switch> enable
Switch# configure terminal
Switch(config)# interface fastethernet 0/6
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 4,5
Switch(config-if)# exit
Switch(config)# exit

Step 4: Configuring Inter-VLAN Routing configuration on Router 5
Router>enable
Router#configure terminal
Router(config)#interface fastethernet 0/0.4
Router(config-subif)#encapsulation dot1Q 4
Router(config-subif)#ip address 192.168.4.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface fastethernet 0/0.5
Router(config-subif)#encapsulation dot1Q 5
Router(config-subif)#ip address 192.168.5.1 255.255.255.0
Router(config-subif)#exit
Router(config)#exit

4.1.3. Assigning IP Adress to each device on VLAN

VLAN 2 (VLAN-A)
PC4:
IP Address: 192.168.2.2
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.2.1
PC5:
IP Address: 192.168.2.3
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.2.1

VLAN 3 (VLAN-B)
PC6:
IP Address: 192.168.3.2
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.3.1
PC7:
IP Address: 192.168.3.3
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.3.1

VLAN 4 (VLAN-C)
PC16:
IP Address: 192.168.4.2
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.4.1
PC17:
IP Address: 192.168.4.3
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.4.1

VLAN 5 (VLAN-D)
PC18:
IP Address: 192.168.5.2
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.5.1
PC19:
IP Address: 192.168.5.3
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.5.1

4.2. Implementing DHCP

4.2.1. DHCP Configuration on Router 1 and Router 6

4.2.1.1. Router 1

Router(config)# interface FastEthernet0/0
Router(config-if)# ip address 192.168.80.1 255.255.255.0
Router(config-if)# no shutdown

Router(config)# ip dhcp pool SUBNET80
Router(dhcp-config)# network 192.168.80.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.80.1
Router(dhcp-config)# exit

4.2.1.2. Router 6

Router(config)# interface FastEthernet0/0
Router(config-if)# ip address 192.168.90.1 255.255.255.0
Router(config-if)# no shutdown

Router(config)# ip dhcp pool SUBNET90
Router(dhcp-config)# network 192.168.90.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.90.1
Router(dhcp-config)# exit

4.2.2. DHCP Configuration on Router 3

4.2.2.1. Router Configuration

Step 1: Configuring fa0/0 with a new subnet
Router(config)# interface FastEthernet0/0
Router(config-if)# ip address 192.168.40.1 255.255.255.0
Router(config-if)# no shutdown

Step 2: Configuring fa0/1 with a different subnet
Router(config)# interface FastEthernet0/1
Router(config-if)# ip address 192.168.50.1 255.255.255.0
Router(config-if)# no shutdown

Step 3: Configuring DHCP for subnet 192.168.40.0
Router(config)# ip dhcp pool SUBNET40
Router(dhcp-config)# network 192.168.40.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.40.1
Router(dhcp-config)# exit
Router(config)# ip dhcp excluded-address 192.168.40.1

Step 4: Configuring DHCP for subnet 192.168.50.0
Router(config)# ip dhcp pool SUBNET50
Router(dhcp-config)# network 192.168.50.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.50.1
Router(dhcp-config)# exit

Router(config)# ip dhcp excluded-address 192.168.50.1

4.2.2.2.Switch Configuration

4.2.2.2.1. Switch 3

Step 1: Configuring interfaces connecting to PCs
Switch(config)# interface range FastEthernet0/3-4
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 1

Step 2: Configuring interface connecting to the Router
Switch(config)# interface FastEthernet0/24
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 1

4.2.2.2.2. Switch 4

Step 1: Configuring interfaces connecting to PCs
Switch1(config)# interface range FastEthernet0/1-2
Switch1(config-if-range)# switchport mode access
Switch1(config-if-range)# switchport access vlan 1

Step 2: Configuring interface connecting to the Router
Switch1(config)# interface FastEthernet0/24
Switch1(config-if)# switchport mode access
Switch1(config-if)# switchport access vlan 1

4.2.3. DHCP Configuration on Router 4

4.2.3.1.Router Configuration

Step 1: Configuring fa0/0 with a new subnet
Router(config)# interface FastEthernet0/0
Router(config-if)# ip address 192.168.60.1 255.255.255.0
Router(config-if)# no shutdown

Step 2: Configuring fa0/1 with a different subnet
Router(config)# interface FastEthernet0/1
Router(config-if)# ip address 192.168.70.1 255.255.255.0
Router(config-if)# no shutdown

Step 3: Configure DHCP for subnet 192.168.60.0
Router(config)# ip dhcp pool SUBNET60
Router(dhcp-config)# network 192.168.60.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.60.1
Router(dhcp-config)# exit
Router(config)# ip dhcp excluded-address 192.168.60.1

Step 4: Configure DHCP for subnet 192.168.70.0
Router(config)# ip dhcp pool SUBNET70
Router(dhcp-config)# network 192.168.70.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.70.1
Router(dhcp-config)# exit
Router(config)# ip dhcp excluded-address 192.168.70.1

4.2.3.2.Switch Configuration

4.2.3.2.1. Switch 5

Step 1: Configuring interfaces connecting to PCs
Switch1(config)# interface range FastEthernet0/1-2
Switch1(config-if-range)# switchport mode access
Switch1(config-if-range)# switchport access vlan 1

Step 2: Configuring interface connecting to the Router
Switch1(config)# interface FastEthernet0/24
Switch1(config-if)# switchport mode access
Switch1(config-if)# switchport access vlan 1

4.2.3.2.2. Switch 6

Step 1. Configure interfaces connecting to PCs
Switch1(config)# interface range FastEthernet0/1-2
Switch1(config-if-range)# switchport mode access
Switch1(config-if-range)# switchport access vlan 1

Step 2. Configure interface connecting to the Router
Switch1(config)# interface FastEthernet0/24
Switch1(config-if)# switchport mode access
Switch1(config-if)# switchport access vlan 1

4.3.Implementing Inter-Network Routing using RIP

4.3.1. Router 1

Router>enable
Router#configure terminal
Router(config)#interface Serial0/2/1
Router(config-if)#ip address 192.169.1.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#router rip

Router(config-router)#version 2
Router(config-router)#network 192.168.2.0
Router(config-router)#network 192.168.3.0
Router(config-router)#network 192.168.80.0
Router(config-router)#network 192.169.1.0
Router(config-router)#exit

4.3.2. Router 2

```
Router>enable
Router#configure terminal
Router(config)#interface Serial0/2/0
Router(config-if)#ip address 192.169.1.2 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface Serial0/2/1
Router(config-if)#ip address 192.169.2.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit

Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#network 192.168.2.0
Router(config-router)#network 192.168.3.0
Router(config-router)#network 192.169.1.0
Router(config-router)#network 192.169.2.0
Router(config-router)#exit
```

4.3.3. Router 3

```
Router>enable
Router#configure terminal
Router(config)#interface Serial0/2/0
Router(config-if)#ip address 192.169.2.2 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface Serial0/2/1
Router(config-if)#ip address 192.169.3.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit

Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#network 192.168.40.0
Router(config-router)#network 192.168.50.0
Router(config-router)#network 192.169.2.0
Router(config-router)#network 192.169.3.0
Router(config-router)#exit
```

4.3.4. Router 4

```
Router>enable
Router#configure terminal
Router(config)#interface Serial0/2/0
Router(config-if)#ip address 192.169.3.2 255.255.255.252
```

Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface Serial0/2/1
Router(config-if)#ip address 192.169.4.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit

Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#network 192.168.60.0
Router(config-router)#network 192.168.70.0
Router(config-router)#network 192.169.3.0
Router(config-router)#network 192.169.4.0
Router(config-router)#exit

4.3.5. Router 5
Router>enable
Router#configure terminal
Router(config)#interface Serial0/2/0
Router(config-if)#ip address 192.169.4.2 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface Serial0/2/1
Router(config-if)#ip address 192.169.5.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit

Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#network 192.168.4.0
Router(config-router)#network 192.168.5.0
Router(config-router)#network 192.169.4.0
Router(config-router)#network 192.169.5.0
Router(config-router)#exit

4.3.6. Router 6
Router>enable
Router#configure terminal
Router(config)#interface Serial0/2/0
Router(config-if)#ip address 192.169.5.2 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit

Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#network 192.168.90.0
Router(config-router)#network 192.169.5.0
Router(config-router)#exit

Router(config)#exit

4.4.Configuring Router Passwords

On Router 1:

Router>enable
Router#configure terminal

Setting the enable secret password
Router(config)#enable secret thapathali

Configuring console line password
Router(config)#line console 0
Router(config-line)#password tha12345
Router(config-line)#login

Configuring VTY line passwords (for remote access)
Router(config)#line vty 0 4
Router(config-line)#password bei2078
Router(config-line)#login
Router(config)#exit

5. VERIFYING THE NETWORK

5.1. Verifying LANs

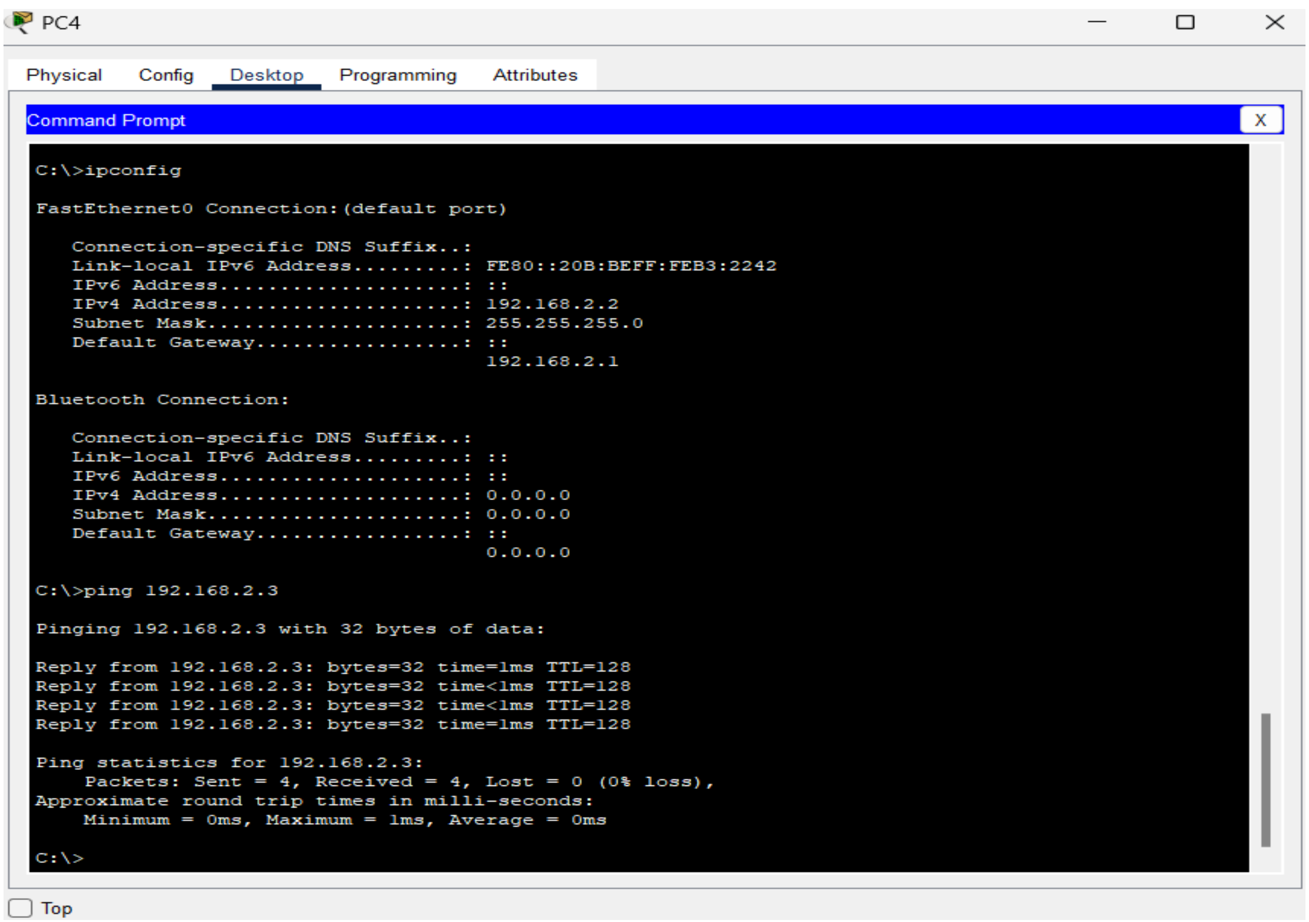


Figure: Sending/Receiving Data Packets from PC4 to PC5 (VLAN-A)

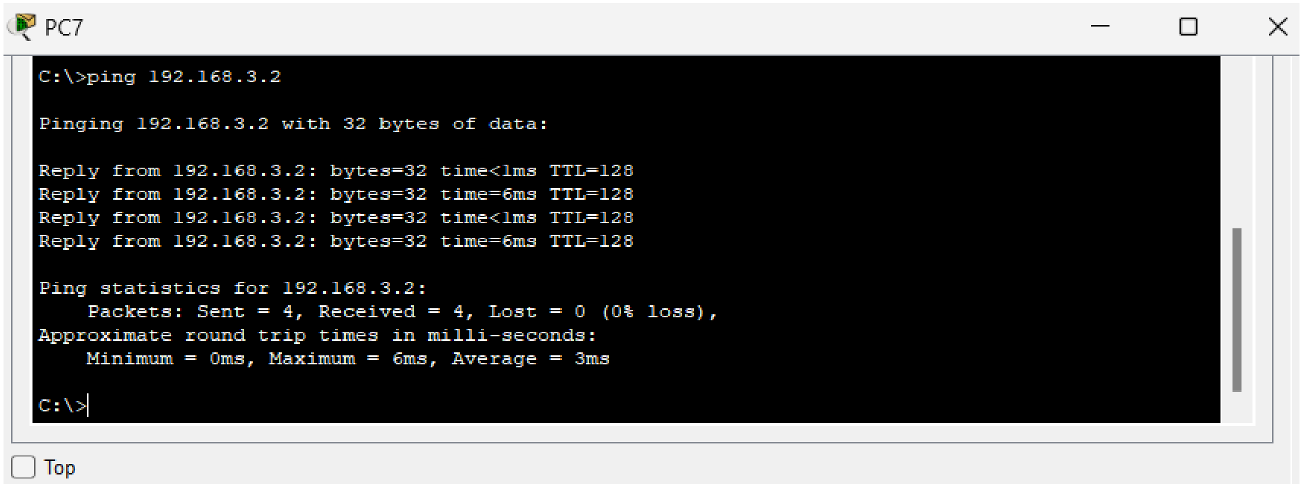


Figure: Sending/Receiving Data Packets from PC7 to PC6 (VLAN-B)

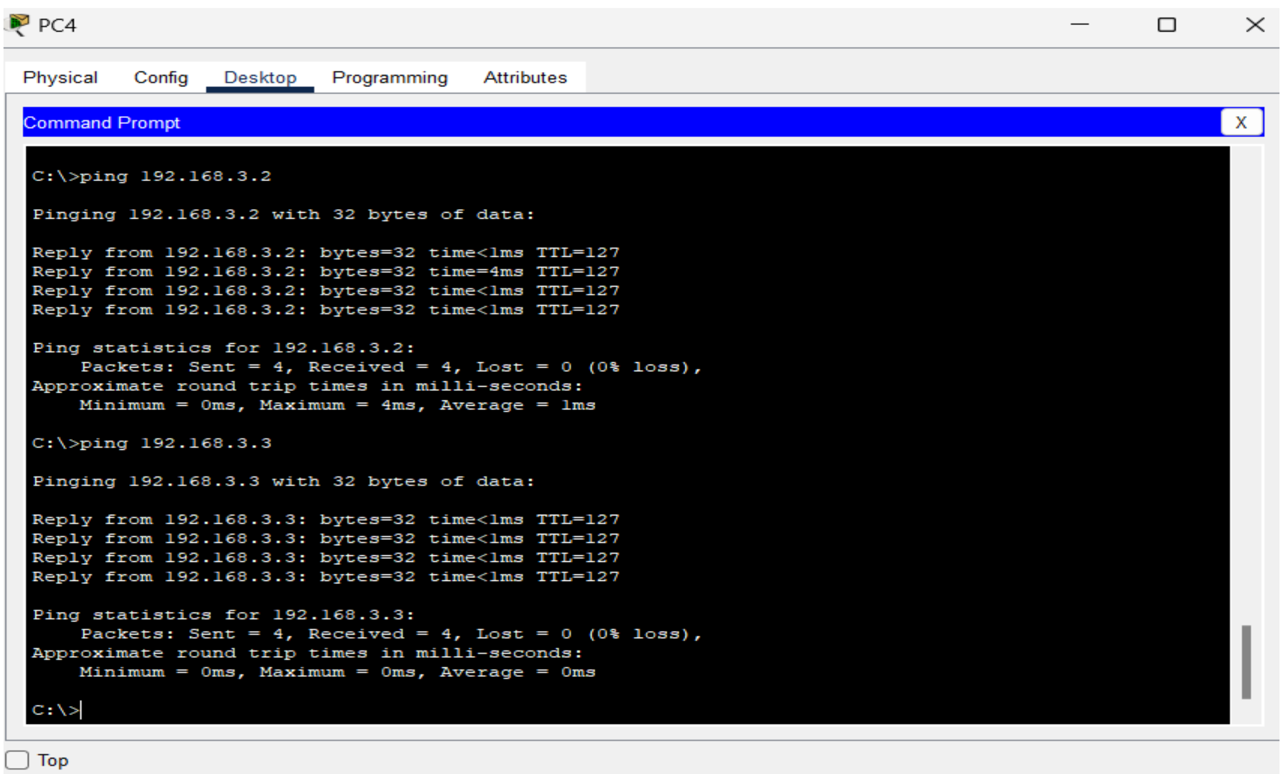


Figure: Sending/Receiving Data Packets from PC4 to PC6 and PC7 (VLAN-A to VLAN-B)

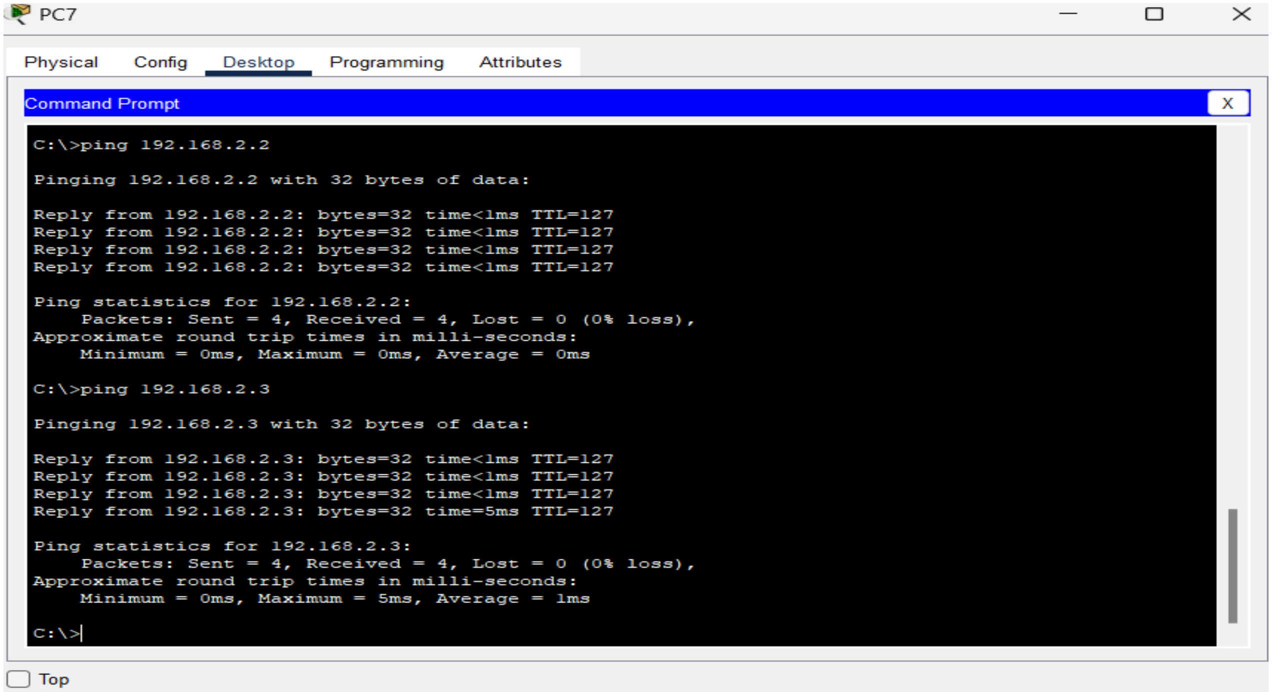


Figure: Sending/Receiving Data Packets from PC7 to PC4 and PC5 (VLAN-B to VLAN-A)

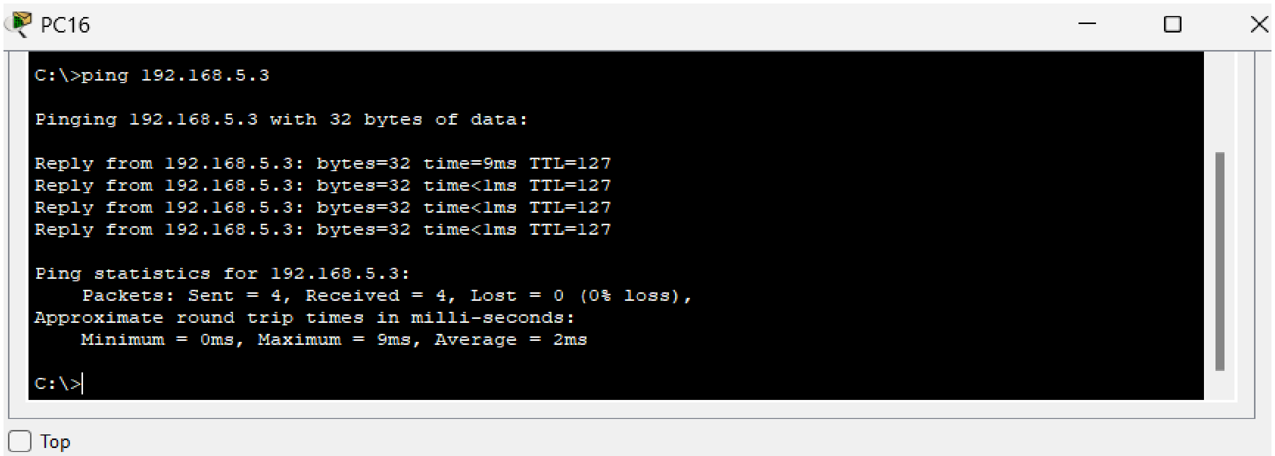


Figure: Sending/Receiving Data Packets from PC16 to PC19(VLAN-C to VLAN-D)

5.2.Verifying DHCP Implementation

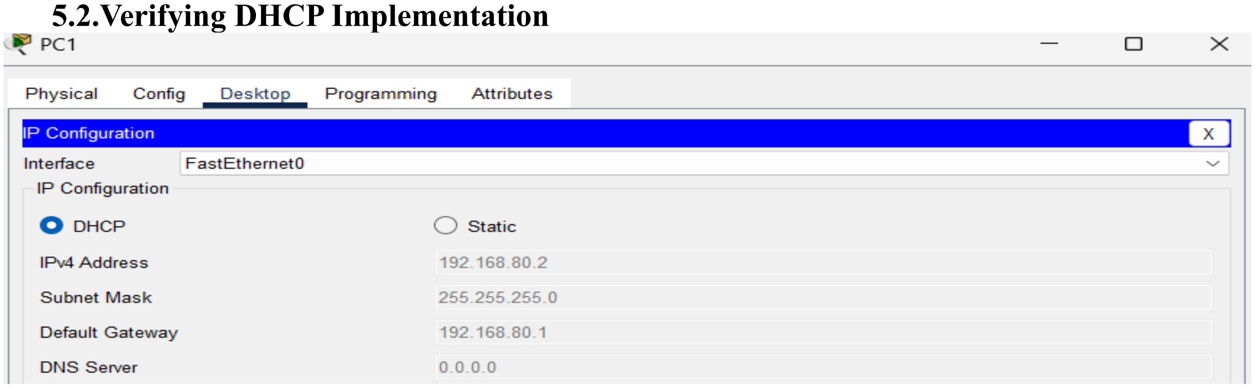


Figure: PC1 got IP address from DHCP Pool created in Router 1

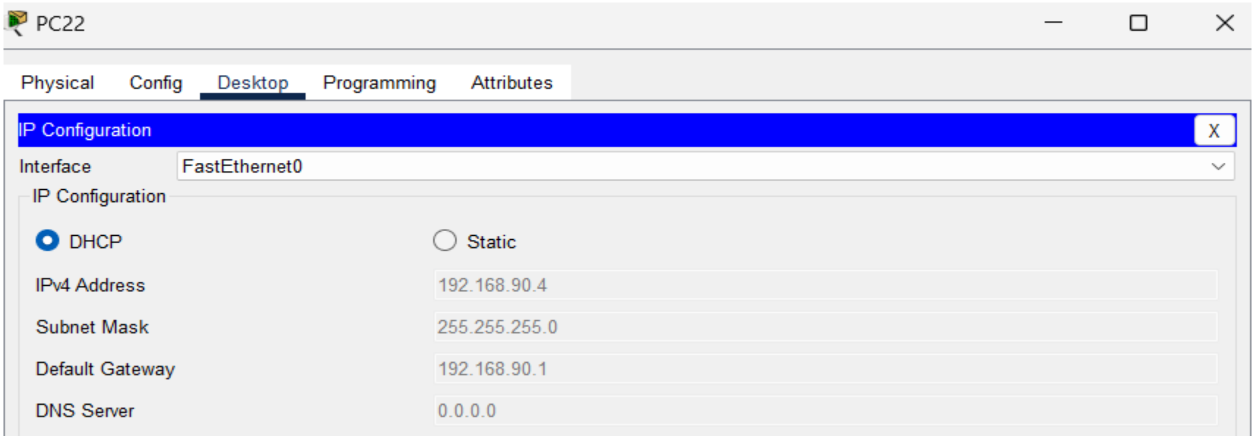


Figure: PC22 got IP address from DHCP Pool created in Router 6

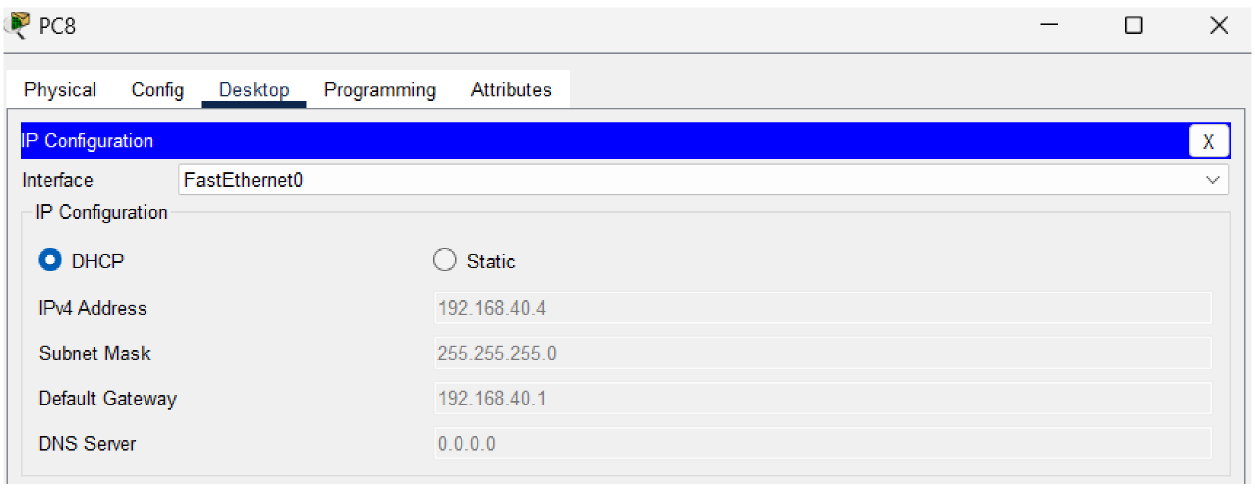


Figure: PC8 got IP address from DHCP Pool-1 created in Router 3

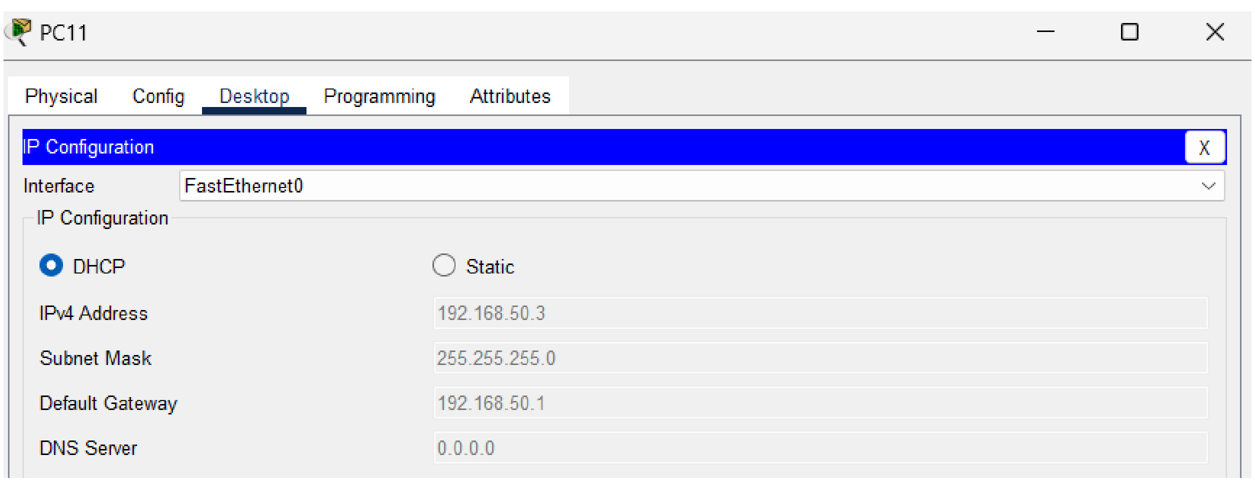


Figure: PC11 got IP address from DHCP Pool-1 created in Router 3

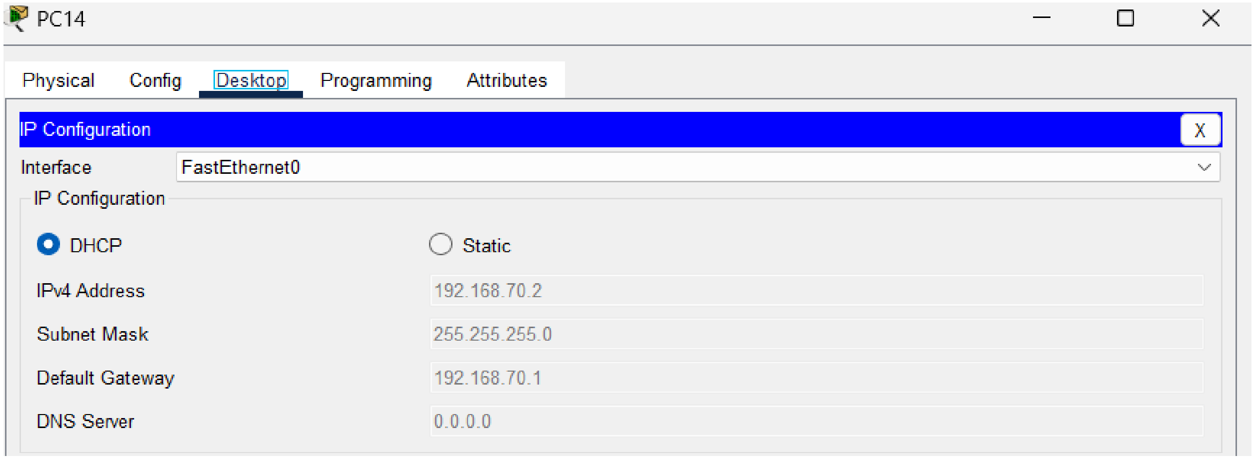


Figure: PC14 got IP address from DHCP Pool-2 created in Router 4

5.3. Verifying Inter-Network Routing

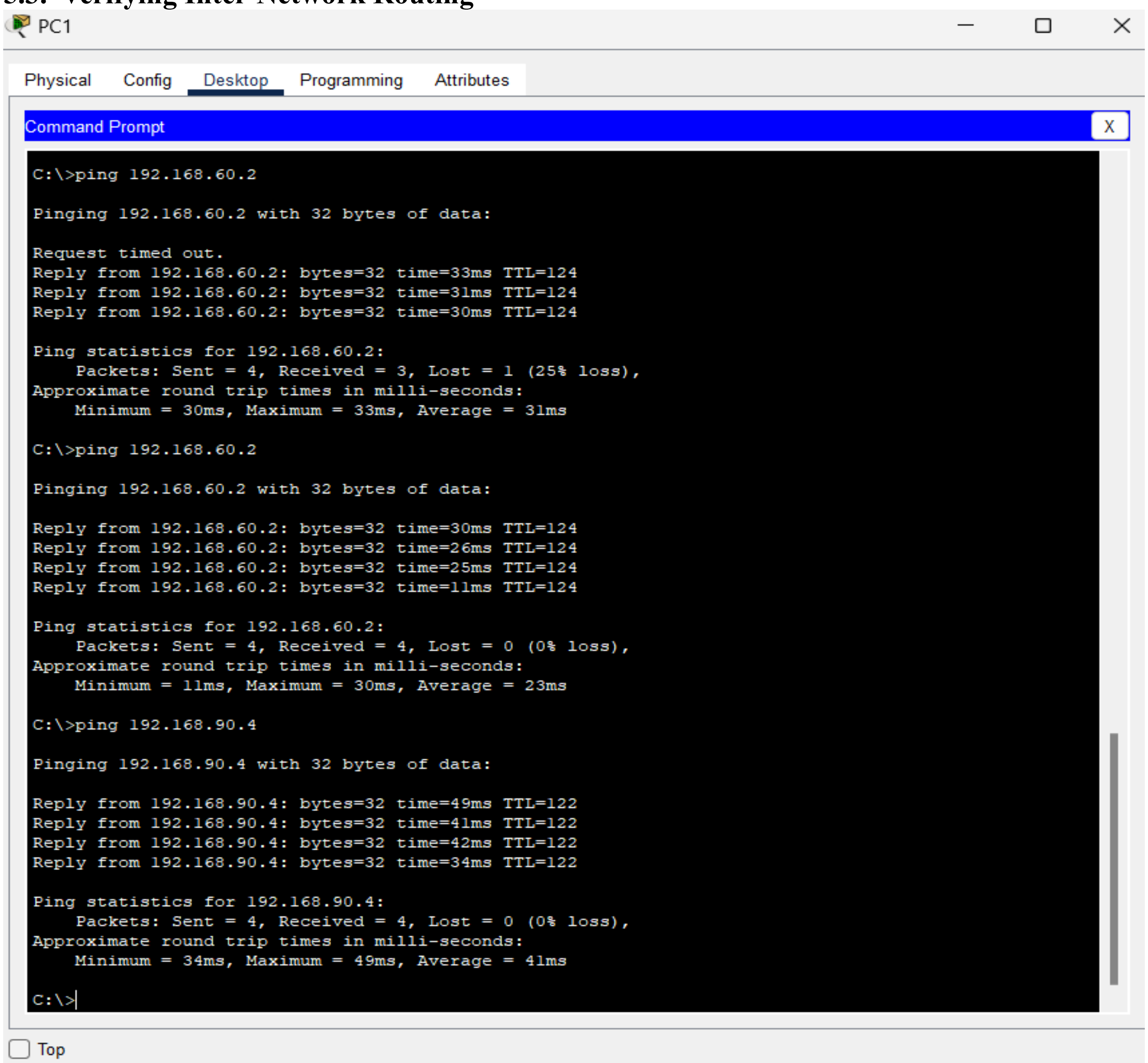


Figure: Sending/Receiving Data Packets from PC1 to PC12 and PC22

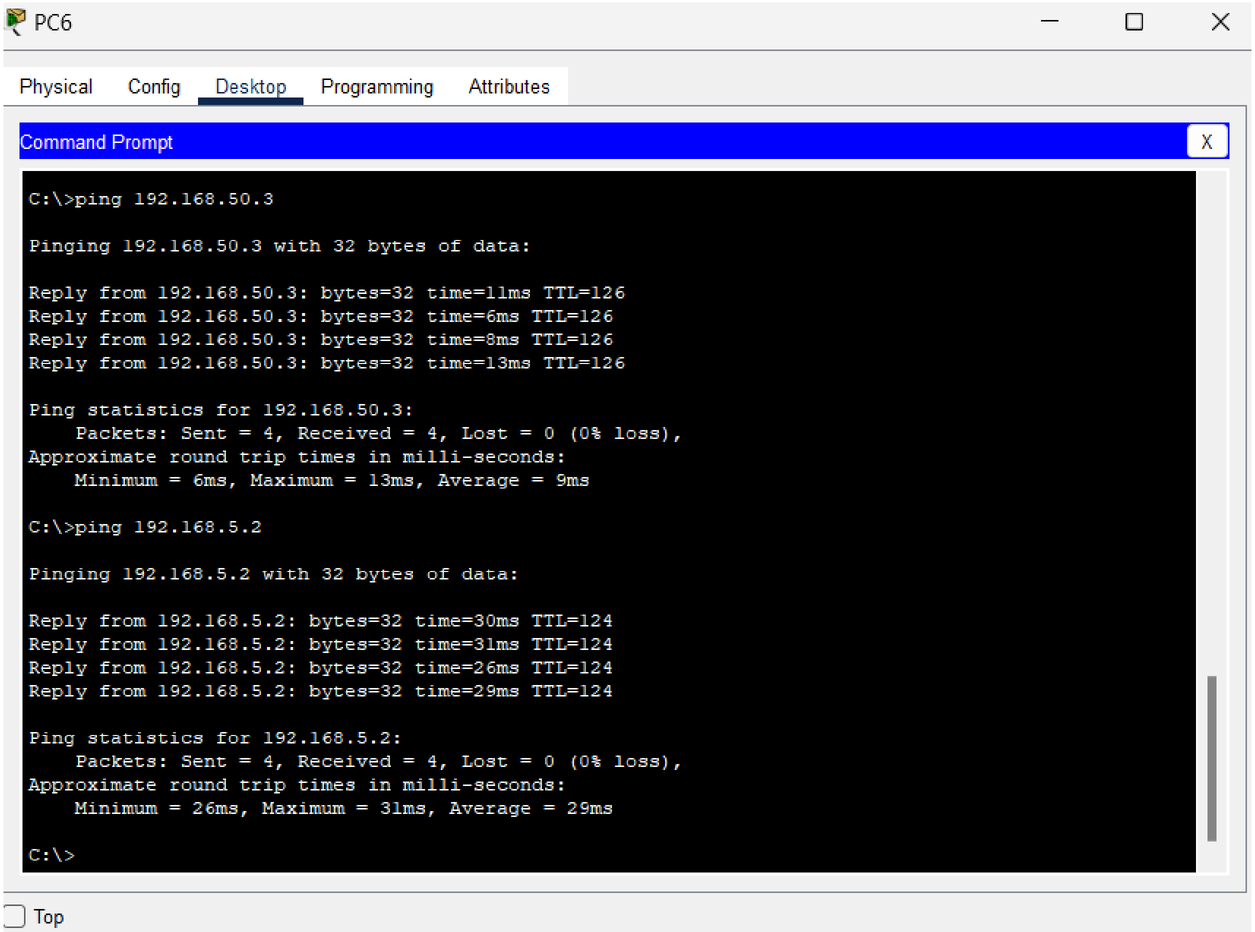


Figure: Sending/Receiving Data Packets from PC6 to PC11 and PC18

5.4. Verifying Router Password

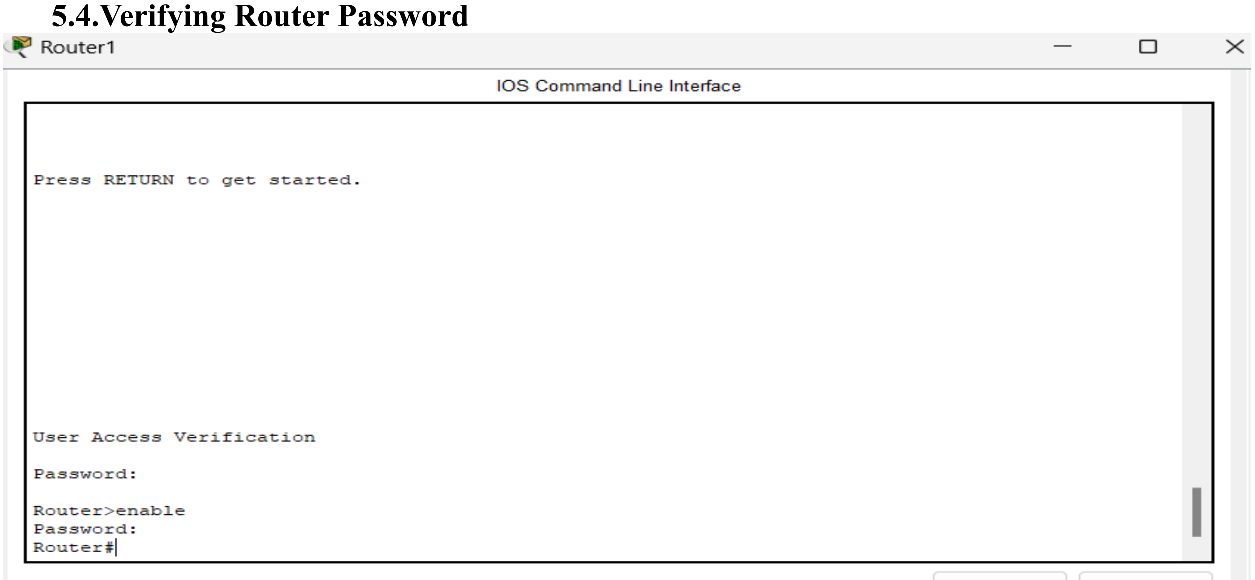
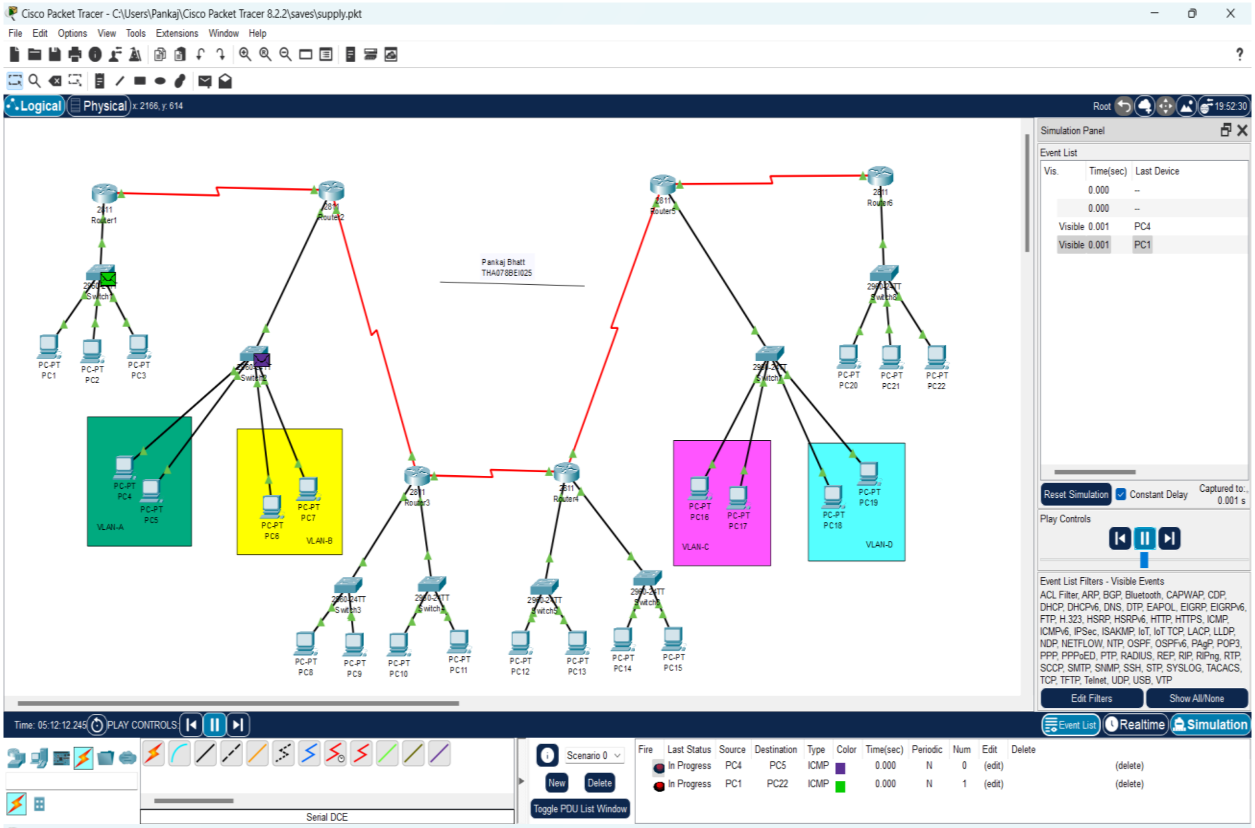
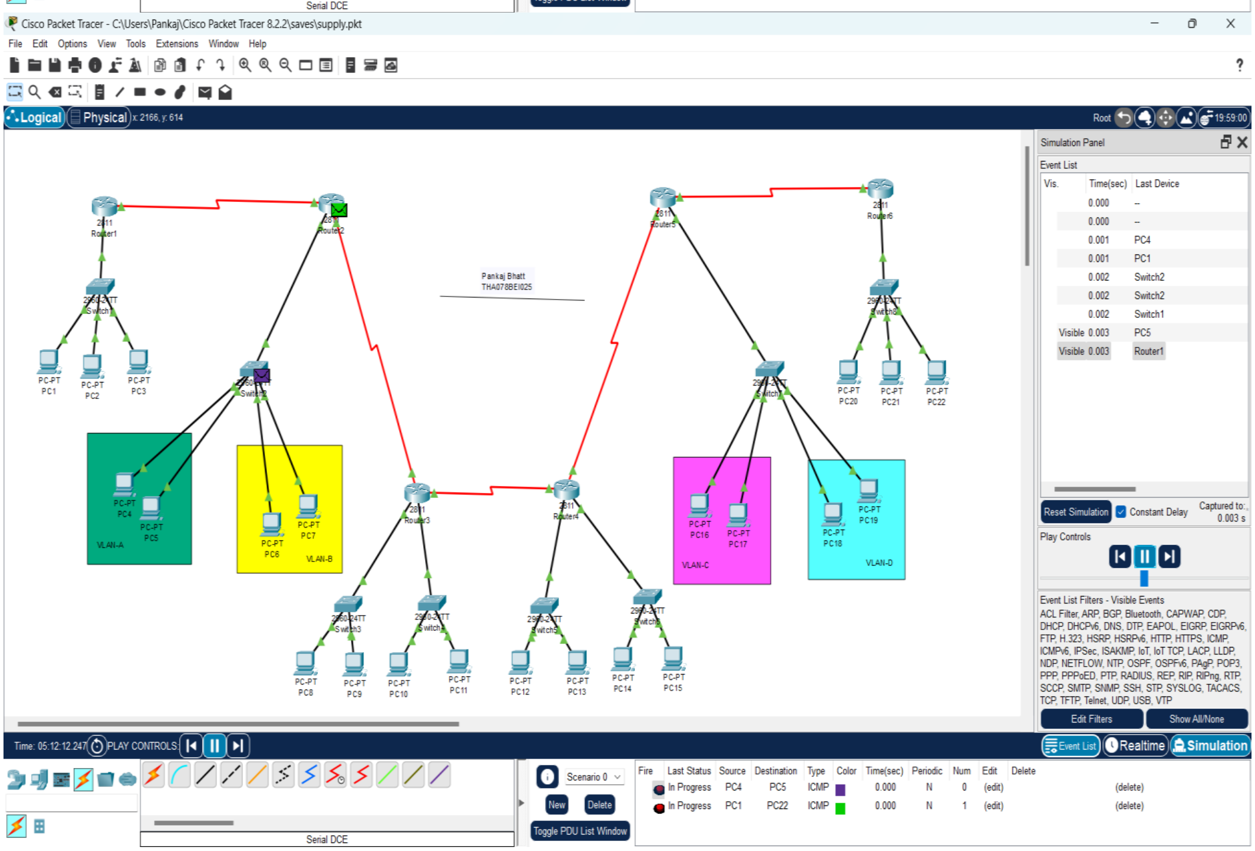


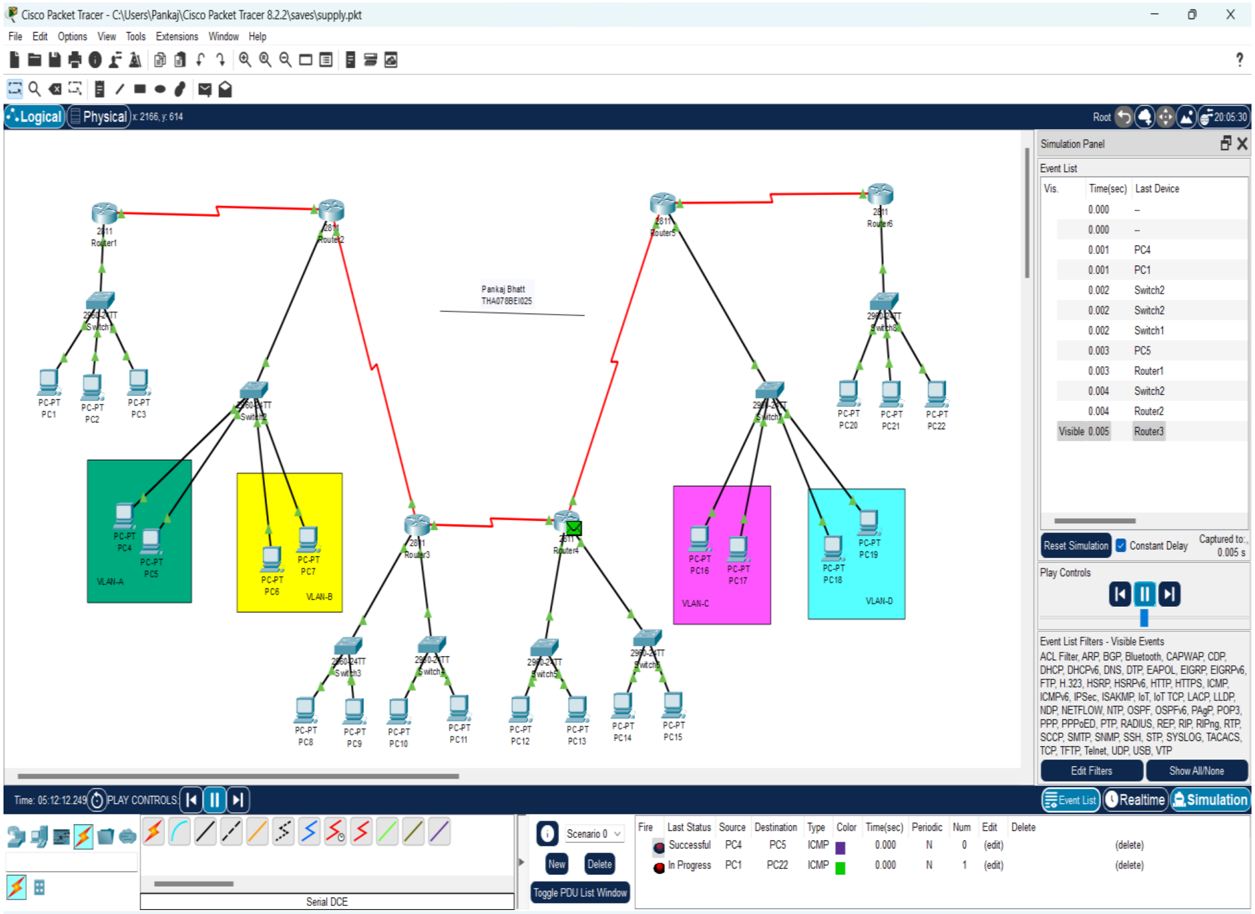
Figure: Password Enabled in Router 1

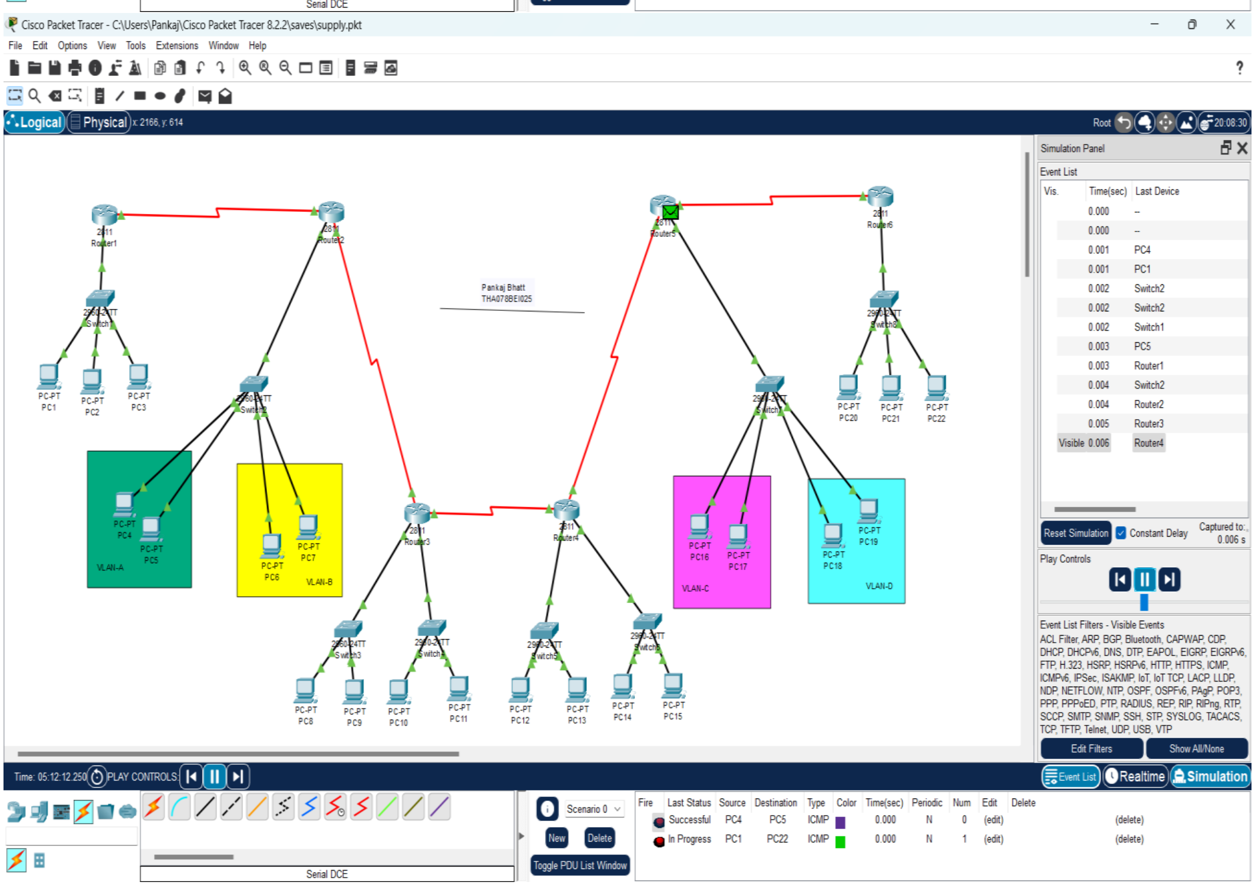
6. SIMULATING THE NETWORK

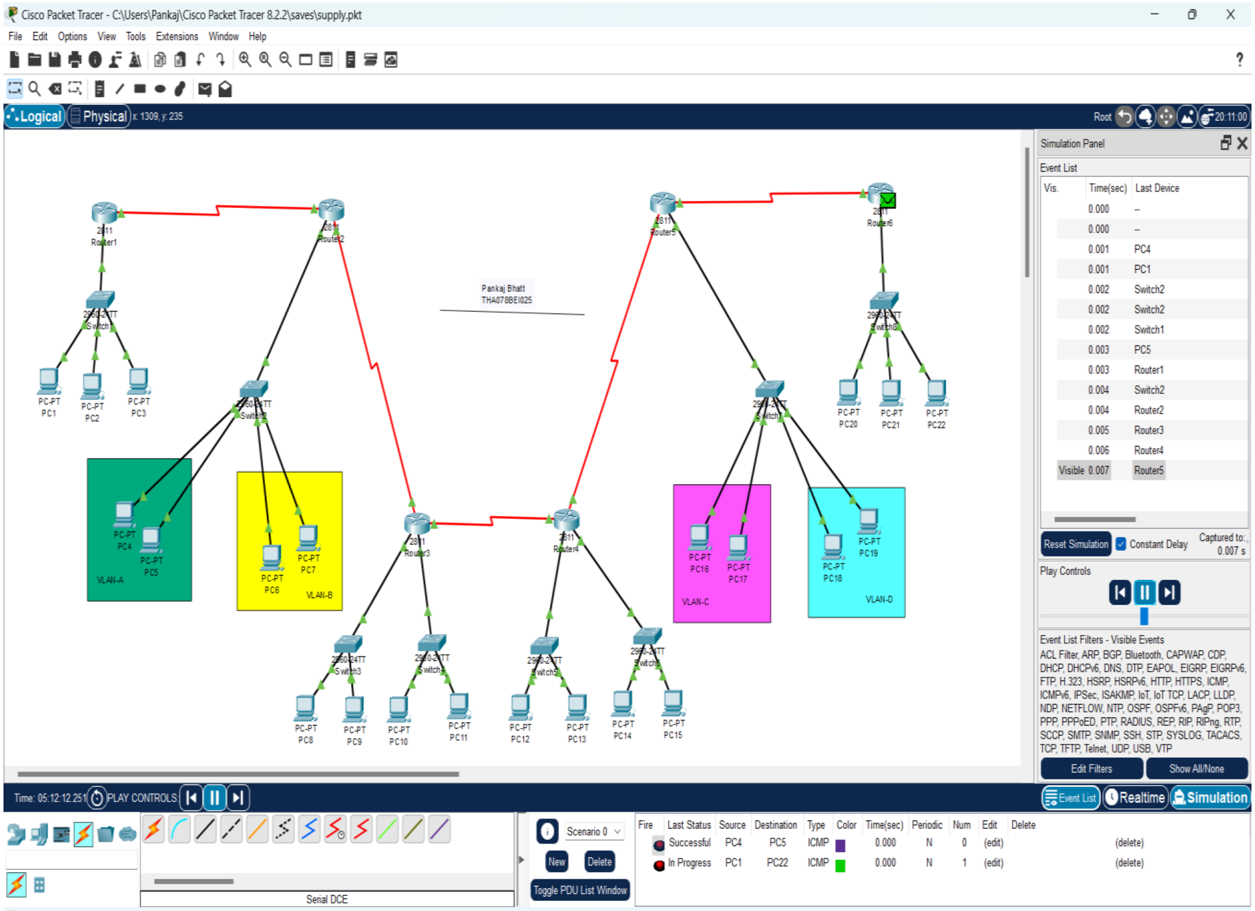
Sending packet from PC1 to PC22

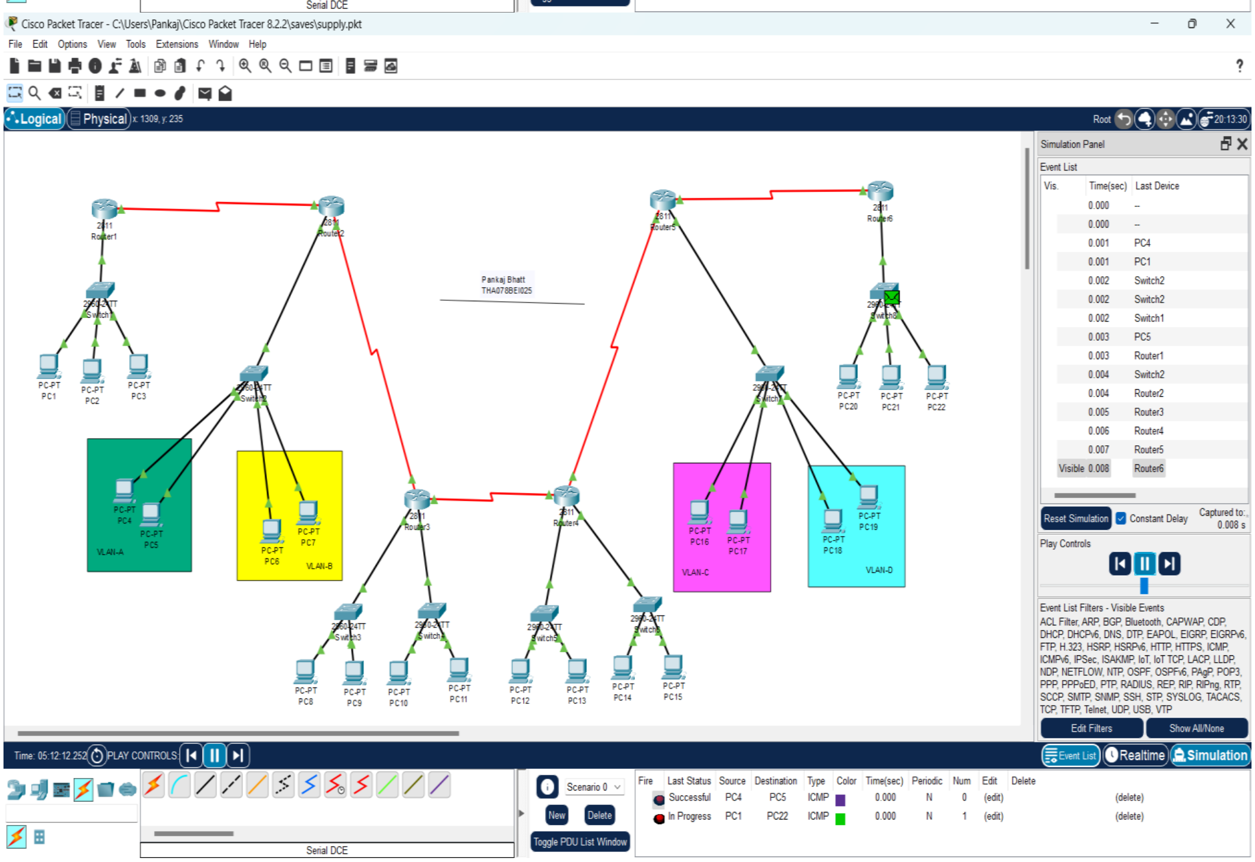


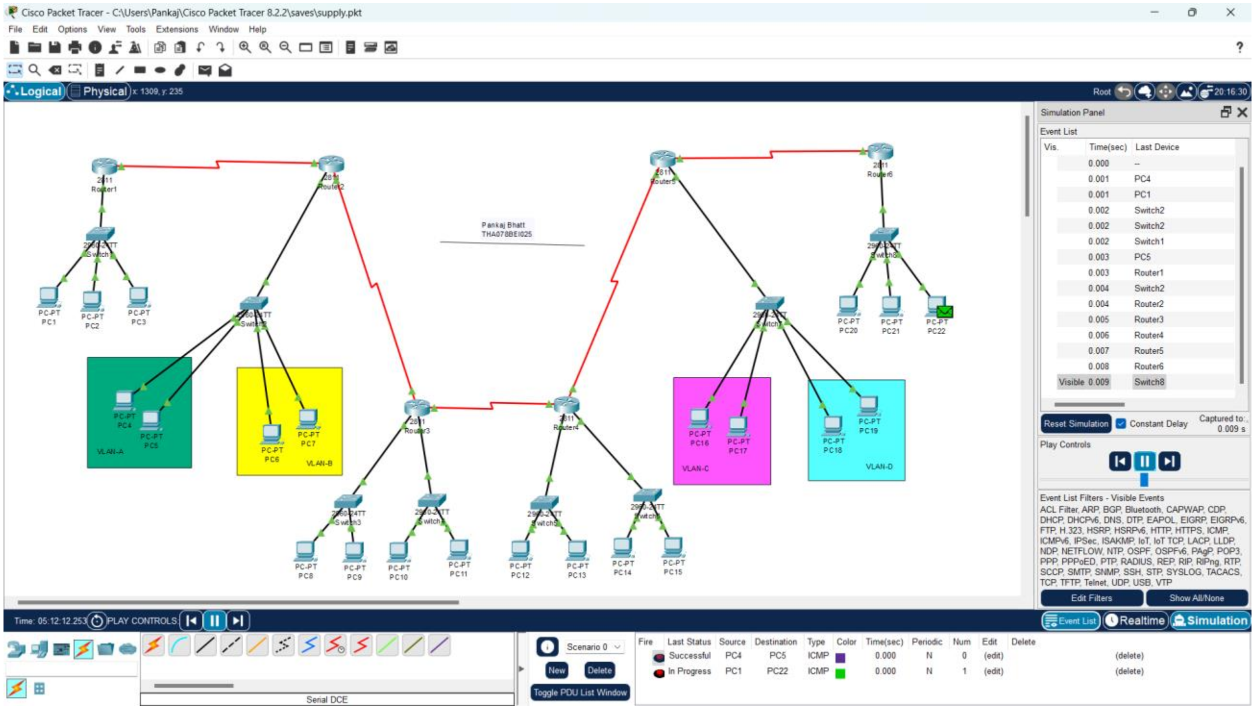



Cisco Packet Tracer - C:\Users\Pankaj\Cisco Packet Tracer 8.2.2\saves\supply.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 2166, y: 614
Root
20:05:30
Pankaj Bhatt
THA078BEI025
VLAN-A
VLAN-B
VLAN-C
VLAN-D
Simulation Panel
Event List
Vis. Time(sec) Last Device
0.000 --
0.000 --
0.001 PC4
0.001 PC1
0.002 Switch2
0.002 Switch2
0.002 Switch1
0.003 PC5
0.003 Router1
0.004 Switch2
0.004 Router2
Visible 0.005 Router3
Reset Simulation
Constant Delay
Captured to: 0.005 s
Play Controls
Event List Filters - Visible Events
ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IPSec, ISAKMP, IoT, IoT TCP, LACP, LLDP, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, POP3, PPP, PPPoED, PTP, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP
Edit Filters
Show All/None
Time: 05:12:12.249
PLAY CONTROLS
Event List
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Serial DCE
Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete
Successful PC4 PC5 ICMP 0.000 N 0 (edit) (delete)
In Progress PC1 PC22 ICMP 0.000 N 1 (edit) (delete)


Cisco Packet Tracer - C:\Users\Pankaj\Cisco Packet Tracer 8.2.2\saves\supply.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 2166, y: 614
Root
20:08:30
Pankaj Bhatt
THA078BEI025
VLAN-A
VLAN-B
VLAN-C
VLAN-D
Simulation Panel
Event List
Vis. Time(sec) Last Device
0.000 --
0.000 --
0.001 PC4
0.001 PC1
0.002 Switch2
0.002 Switch2
0.002 Switch1
0.003 PC5
0.003 Router1
0.004 Switch2
0.004 Router2
0.005 Router3
Visible 0.006 Router4
Reset Simulation
Constant Delay
Captured to: 0.006 s
Play Controls
Event List Filters - Visible Events
ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IPSec, ISAKMP, IoT, IoT TCP, LACP, LLDP, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, POP3, PPP, PPPoED, PTP, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP
Edit Filters
Show All/None
Time: 05:12:12.250
PLAY CONTROLS
Event List
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Serial DCE
Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete
Successful PC4 PC5 ICMP 0.000 N 0 (edit) (delete)
In Progress PC1 PC22 ICMP 0.000 N 1 (edit) (delete)







7. CONCLUSION

This computer network project has successfully demonstrated the critical importance and intricate workings of modern networking technologies. By designing, implementing, and testing various network components including LANs, VLANs, and dynamic routing protocols such as RIP, alongside essential configurations like DHCP, I have gained a comprehensive understanding of how these elements interact to create a robust, efficient, and secure network infrastructure. The project's success underscores the value of computer networking as a fundamental pillar in our digital age, enabling seamless communication, efficient data management, and enhanced security. This endeavor has not only enriched my technical skills but also highlighted the transformative impact of network technologies on both small-scale and large-scale environments. I extend my appreciation to the field of computer networking for its pivotal role in connecting the world and driving technological progress

I would like to thank my subject teacher Mr. Sharad Chandra Joshi for providing this opportunity to learn in depth about network implementation in the form of this project.